

INORGANIC CHEMISTRY LABORATORY

Qualitative Inorganic Analysis

CHE 1204

Department Of Physical Sciences

Faculty Of Applied Sciences

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SAFETY IN THE LABORATORY

Safety in the laboratory must be emphasized. The chemicals you will work have some hazards associated with them. Therefore, it is important to follow the safety rules outlined in this lab manual. You should assume that all compounds encountered in the laboratory are toxic and handle them carefully. Safety goggles for eye protection are recommended and lab coats, gloves and face masks are to be worn by all students at all times when entering the laboratory. Many chemicals, common in chemical laboratories, will make damages in clothing. Always wash your hands thoroughly when leaving the laboratory. You should become familiar with the proper use of the safety shower, eye-wash fountain, fire blanket and fire extinguisher.

Report any accidents which occur immediately to the Demonstrator/ Technician.

The following safety rules to be strictly followed by all students.

- Everyone must wear a lab coat during lab. Wearing goggles/safety glasses, gloves, face masks are highly recommended. Students who come to lab without proper clothing and shoes will be asked to go back for change. If they do not come on time it will be counted as an absence.
- Do not touch chemicals with your hands. Spatulas will be provided for handling solid materials. Do not taste any chemical. Do not smell any chemicals directly. Use your fingers to waft the odor to your nose.
- Long hair should be securely tied back to avoid the risk of setting it on fire.
- Do not pipette solutions by mouth. Pipette fillers are provided at each lab station.
- Do not put flammable liquids near an open flame. When finished with your Bunsen burner for a given portion of an experiment, turn it off.
- When heating a test tube, make certain that the open end of the tube is directed away from you.
- Do not sit on the lab benches. Do not leave your coats and back packs on the benches.
- Use of mobile phones are strictly prohibited in the laboratory.
- Do not pour any chemicals into a sink without authorization from the demonstrator. There will be proper waste containers in the lab. Notify your demonstrator /lab staff if a mercury spill should occur.
- All broken glassware should be cleaned. The demonstrator should be notified immediately.
- Do not spill any chemical directly on the balance pan. Always use weighing boats.

- If a chemical gets on your skin, immediately wash the affected area with enough quantities of water/immediately remove the contaminated clothing and use the safety shower. The instructor should be notified no matter how insignificant the incident might seem.
- When pouring one liquid into another, do very slowly and cautiously. To dilute an acid, pour the acid into the water, never pour water into an acid.
- No student shall be permitted to work alone in the lab without permission, you should be supervised by a laboratory demonstrator.
- Exercise good housekeeping practices in the laboratory, Be sure that the lab benches remain free of disorder during the experiment. In the event of a spill, clean the area immediately.
- Know what you have to before entering the lab. Read the experiment carefully before coming to the laboratory.

Safety equipments

- Become familiar with the location and operation of the nearest fire extinguisher.

Eye Protection

- Safety goggles must be worn at all times during any laboratory activity that is potentially harmful to the eyes

Protective Clothing

- Clothing worn in the lab should cover as much of the body area.
- Always wear a lab coat and avoid wearing loose cloths. Rubber slippers are prohibited.
- Gloves are used for protection against skin contact or cuts when handling certain chemicals, hot or cold objects, or glass.

Respiratory Protection

- Fume hoods are used for work involving hazardous or malodorous materials

Fire Aid Equipment

- Safety Showers are used for large chemical spill contact on a person.
- Eye Wash stations are used if something enters the eye. The eye should be held open and flushed for 10 – 15 minutes. After flushing seek medical attention.

Common Laboratory Equipment



Techniques in Qualitative Inorganic Analysis

Explained below are some fundamental techniques that are essential when carrying out qualitative analysis.

Adding reagents

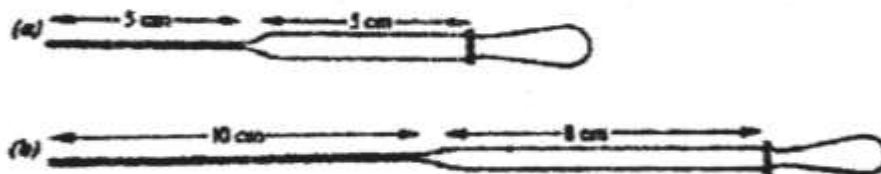


Figure 01: Droppers

- Before using the dropper, wash it thoroughly at least twice with distilled water.
- Then, add the reagent drop-wise. The dropper should be washed thoroughly at least three times after using it.
- When it is not in use, store the dropper in a beaker of clean water.
- Care must be taken to ensure that no liquid enters in to the bulb of the dropper.

Reagent bottles

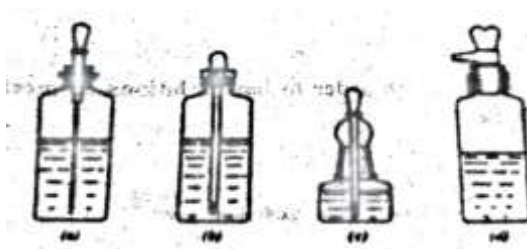


Figure 02: Different types of Reagent bottles

- Reagent bottles and their stoppers should not be put on the bench. They should be returned to their correct places in the shelves immediately after use.
- Use a dropper to transfer reagents from one reaction tube to another and never pour the liquid directly from the reagent bottle.
- Use the minimum quantity of the required reagent. Adding excess reagents increases the volume of the solution and causes experimental errors.
- The remaining reagents should not be poured back into the reagent bottle if the dropper does not belong to the reagent bottle used.

Pouring reagents from reagent bottles

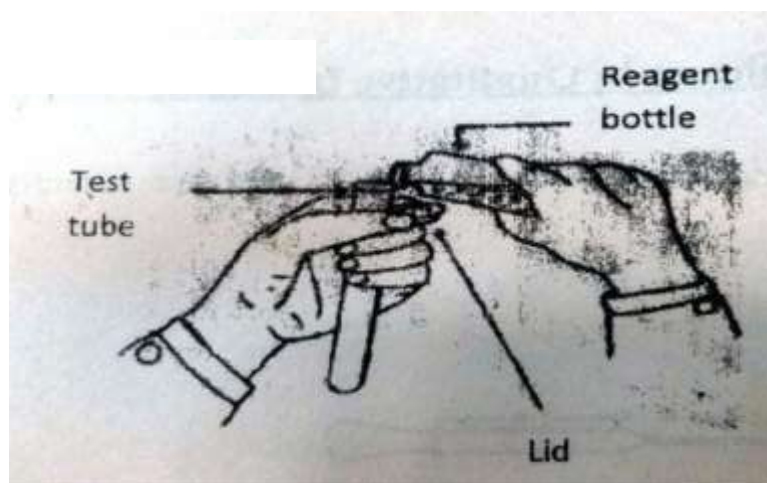
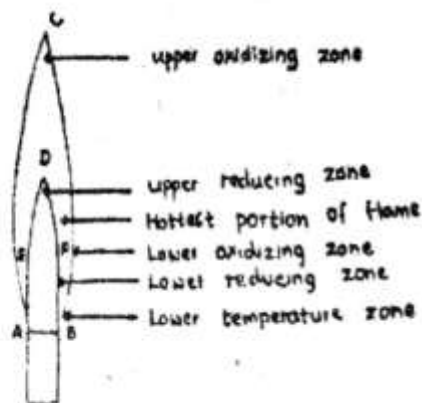


Figure no: 3 -pouring reagents

- When pouring reagents from bottles, the test tube into which the reagent is to be poured is held with three fingers of the left hand as shown.
- The reagent bottle from the rack is held with the right hand.
- The bottle stopper is removed with the left thumb and index finger and the reagent is poured into the test tube while holding the test tube at an angle.
- After transferring the reagent, the stopper is replaced, and the reagent bottle is kept back in the side-rack.
- The reagent bottle should not be on the bench at any time, which would prevent accidents due to knocking over the bottle.

Heating a solution in a test tube



- A Bunsen burner is employed to heat solutions. In order to heat solutions correctly you should know the principle parts of the flame.

Figure 04: Parts of the Bunsen burner flame

By adjusting the amount of air that is mixed with gas (using the air hole at the bottom part of the burner body), different types of flames may be obtained for different purposes.

- I. A Fully open air hole gives a non-luminous (blue) flame.
- II. A Fully closed air hole gives a luminous (yellow) flame.
- III. A partially open air hole gives a flame consisting of an inner blue cone with luminous tip and an outer non-luminous zone.



Figure 05: Heating a solution

- The solution to be heated is poured in to a test/ignition tube. The tube is held with the test tube holder/a pair of tongs, and the bottom part of the tube is heated in the yellow part of the flame while constantly moving the tube in a circular (gentle swirling) motion.
- This ensures uniform distribution of heat in the tube, which prevents cracking of the tube and also overheating of solutions (resulting in splattering)
- The mouth of the tube should be pointed away from people nearby.
- For most types of heating, a small-medium flame is sufficient. The size of the flame may be controlled by adjusting the gas flow.

Separation of a supernatant liquid from a preparation

- Centrifuge the solution.
- Take the test tube out from the centrifuge and hold it at an angle (45°) in the left hand.
- Compress the rubber teat or nipple of the dropper and insert it into the tube until it is just below the liquid level.

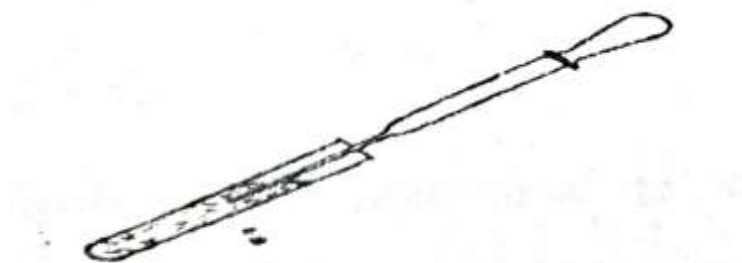


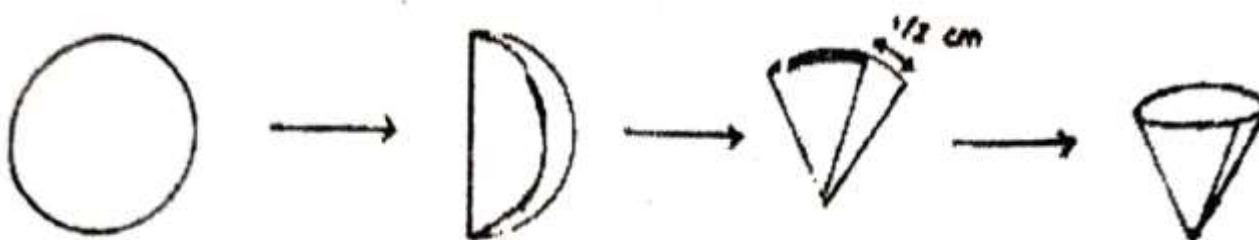
Figure 06: Separation of a supernatant using a dropper

- Release the pressure of the teat slowly so that the liquid enters the dropper.
- Note: Never release the pressure suddenly.
- Great care should be taken to ensure that the tip of the dropper does not touch the precipitate. The solution in the dropper should be perfectly clear.

Filtering

- Filtering is a technique used in macro analysis, not in semi micro analysis. In semi micro analysis, centrifugation followed by decantation is used instead of filtering.

Method 01



Method 02

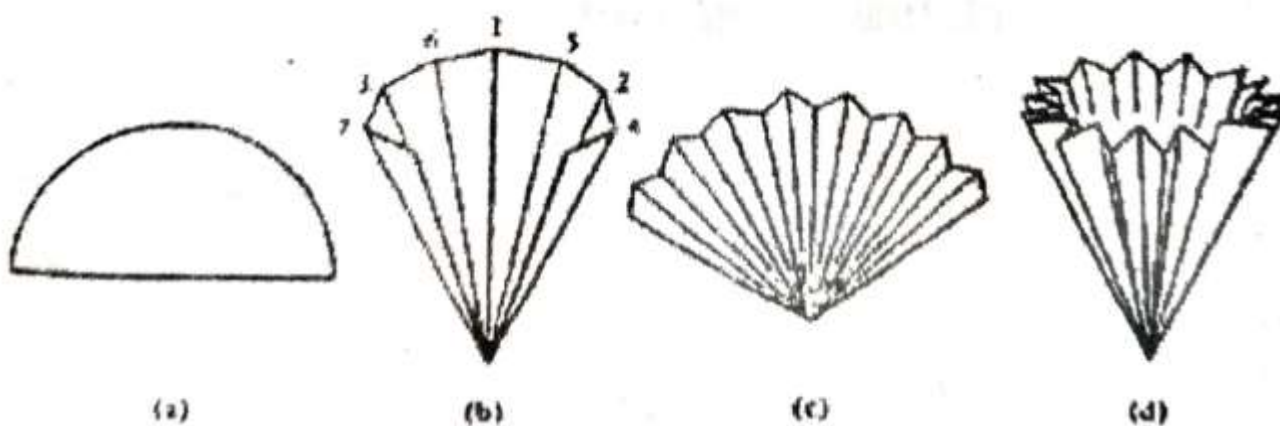


Figure 07: How to fold a filter paper

- Filter paper can be fold using any of the above two methods.
- It is next placed well set against the walls of the funnel and moistened with distilled water. Care should be taken to prevent any air getting trapped between the filter paper and the walls of the funnel.
- The solution and the precipitate that needs to be filtered are poured with the aid of a glass rod into the funnel.
- After pouring the solution, the test tube from which it was poured is washed with distilled water (a few times) and the washings are also poured along the glass rod as before.
- Finally, the glass rod is also washed using distilled water from the wash bottle.

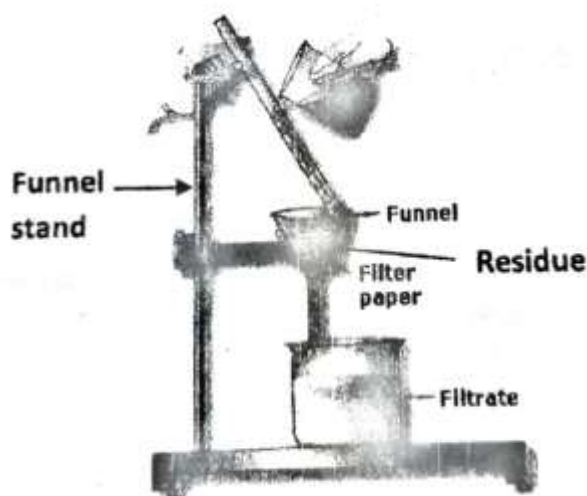


Figure 08: Filtration of a solution

Gas handling

- The apparatus required for gas delivery is shown below.
- Test tube 1 containing the mixture emitting the gas to be delivered, with the delivery tube connected to it, is first heated until gas emission can be clearly seen from the tip of the gas delivery tube.
- Test tube 2 is placed in such a manner as to allow this gas to purge through the solution contained in it.

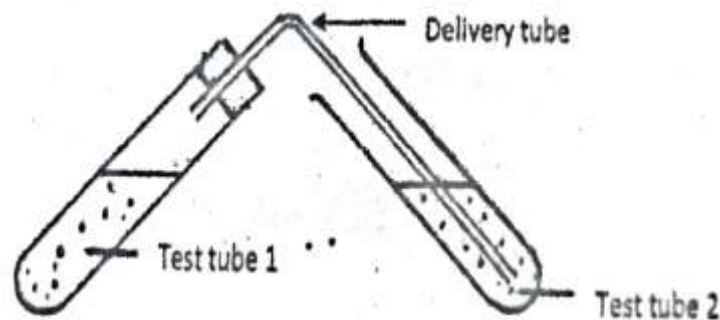


Figure 09: Apparatus for gas delivery

- A sufficient volume of solution should be present in test tube 2 to allow the bottom part of the gas delivery tube to be immersed in it.
- During the heating step, care should be taken to move the whole arrangement in a circular motion to avoid overheating the solution in test tube 1 and also to provide a uniform heating rate.
- The cork around the short arm of the delivery tube should fit test tube 1 snugly without any leaks.
- When purging is complete, test tube 2 should be removed first and thereafter test tube 1 is removed away from the flame. Otherwise, the vacuum created when the solution in test tube 2 cool down away may result in the solution in test tube 1 being sucked into test tube 2, and undesirable consequences may occur.

Evaporating a solution to dryness

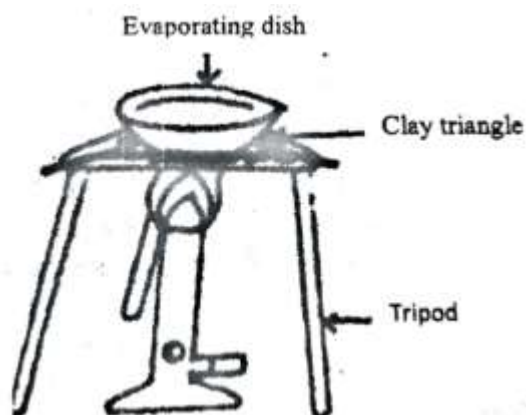


Figure 10: Evaporating a solution

- The solution is placed in an evaporating dish, and the dish is placed on the clay triangle kept on the stand as shown.

- Heating is first done with a flame to avoid splattering of the contents in the evaporating dish.
- The size and the temperature of the flame may be gradually increased as the evaporation progresses if needed.
- In many cases, if the flame is removed while a small amount of liquid still remains in the evaporating of the remaining liquid.

Systematic analysis on the Semi micro scale

Follow the sequence given below in semi micro scale for the analysis of the sample.

1. Physical Characterization
2. The preliminary examination of the solid sample.
3. Testing for anions in solution.
4. Testing for cations ion solution.

Physical Characterization

- 1) Appearance: Record the of the substance carefully. Observe whether it possesses any characteristic colour or odor and record them.
- 2) Solubility: Many of the inorganic compounds that will be encountered in this course will be solids. It is useful to find out their solubility in a solvent. It is essential for the compound to be in solution before the qualitative analysis of anions and cations.

Solubility is determined by taking a small amount (5-10 mg) of the material in a test tube and adding a solvent, followed by vigorous shaking. Solubility should be tested in following solvents in the order given below.

1. Distilled water (cold followed by hot)
2. Dilute HCl (cold followed by hot)
3. Conc HCl
4. Dilute HNO₃
5. Conc.HNO₃
6. Aqua regia (Conc.HCl:Conc.HNO₃ v/v ratio)

Note: For water soluble samples only, record the action on litmus papers

Preliminary Tests

1. Flame Test

Certain metal salts, when heated in a flame, give characteristic colours to the flame. This test is carried out by using a platinum or nichrome wire mounted on a glass rod for easy handling.

Carry out the flame test for 2-3 mg of the mixture

The wire is first cleaned thoroughly by dipping in a test tube containing conc.HCl and heating strongly in the colourless or pale blue region of an oxidizing flame until various colours resulting from impurities cease to appear. At this point, the wire should have a dull red glow.

Next, the wire is dipped in a solution of the substance in conc.HCl (prepared by placing a small amount of the substance being tested on a watch glass and adding a few drops of conc HCl) and heated again in the oxidizing flame. The wire should first be brought close to the flame when more volatile chlorides impart colours to the flame, and thereafter heated in the hottest part of the flame.

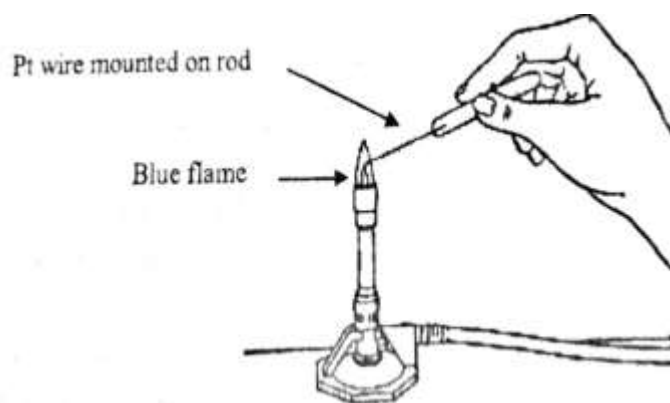


Figure 11: Flame test

After the test is complete, the wire is allowed to cool and then cleaned with conc.HCl as explained above before storing or carrying out another test. Unless it is cooled, the glass rod on which the wire is mounted may disintegrate when the (hot) glass rod makes contact with the (cold) HCl.

Observation	Inference
Persistent golden-yellow flame	Na^+
Violet flame	K^+
Brick – red flame	Ca^{2+}
Crimson flame	Sr^{2+}

Yellowish –green flame	Ba^{2+}
Livid-blue flame	$\text{Pb}^{2+}, \text{As}^{3+}, \text{Sb}^{3+}, \text{Bi}^{3+}, \text{Cu}^{2+}$
Red	Li^{+}

Note: The Na flame makes that of other elements, e.g. that of K. Therefore, view the flame through two thickness of cobalt blue glass, whereby the yellow colour due to Na is mask or absorbed and the other colours are modified as listed below.

Observation through cobalt glass	Inference
Nil	Na^{+}
Crimson	K^{+}
Light green	Ca^{2+}
Purple	Sr^{2+}
Blue green	Ba^{2+}

2. Heating in a charcoal block

This involves placing a small amount of the substance being tested inside a small depression dug into a charcoal block (by scraping of the surface of the charcoal block using a sharp object like a spatula end.) Wetting it with some distilled water using a dropper (to prevent the substance from being dislodged), and then heating the substance in oxidizing and reducing flames.

A small luminous flame (i.e. with the air hole completely closed) 2-3cm in height should be used in these tests. To obtain a reducing flame, the blow pipe should be placed outside this flame and blown gently to turn the inner core of the flame towards the charcoal block so that the substance gets heated by the inner part of the flame. An oxidizing flame can be obtained by keeping the blowpipe inside the flame and blowing air through it so that outer cone of the flame reaches the substance. Another useful test is to add a drop of cobalt nitrate to the substance and heating on the charcoal block.

In the first test, the appearance of beads incrustated with metal may indicate the presence of certain metals. At times substances that are difficult to decompose upon heating are encountered. In such cases, a sodium carbonate flux (which is actually a mixture of Na_2CO_3 and K_2CO_3 in the ratio of 1:1). This resultant mixture can undergo reactions more readily and may be tested as above. When cobalt nitrate is added to

the substance and heated in a charcoal block, differences in colours may give an indication as to what may be present in the substance tested.

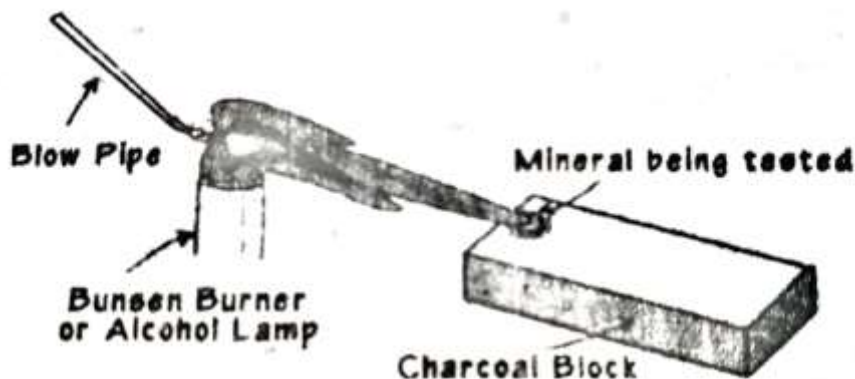
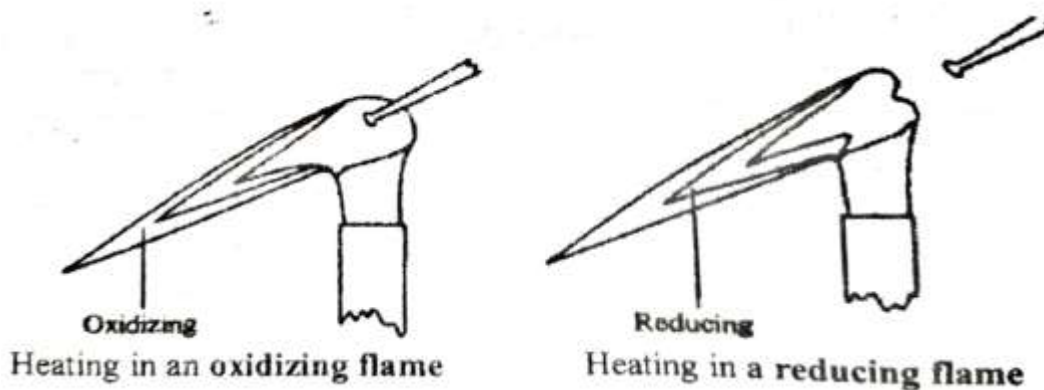


Figure 12: Charcoal Block Test



Carry out following procedures:

- I. Carry out charcoal block for 2-3 mg of the mixture.

If the substance deflagrates, oxidizing agents (NO_3^- , NO_2^- , ClO_4^- , IO_3^- , MnO_4^-) may be present. Brown incrustation indicates the presence of Cd^{2+} . If the substance fuses and is absorbed by the charcoal, or forms a liquid bead, Salts of alkalis and alkaline earths may be present.

- II. MIX 2-3 mg with twice its bulk of anhydrous Na_2CO_3 , place it in a charcoal block and heat in the reducing flame.

Observation	Inference
White incrustation, Yellow when hot	ZnO
Brown incrustation	CdO
White incrustation, brittle metal	As ₂ O ₃
Brittle yellow incrustation	Bi
Yellow incrustation, malleable and marks paper	Pb

III. Moisten the residue of experiment (II) with one or two drops of Cobalt nitrate and ignite strongly.

Observation	Inference
Blue residue	Al ₂ O ₃ , Phosphates, Arsenates, Silicates, Borates
Green residue	ZnO
Pink residue	MgO

3. Borax Bead Test

A small loop is made at the end of the Platinum or Nichrome wire, cleaned as stated before, heated, and dipped in some borax placed on a watch glass to pick up some solid borax. This solid is then heated with constant turning of the wire to prevent molten borax from falling off. A colourless bead of borax is obtained during this process. The borax solid expands on heating before it fuses and forms a bead, and heating should be continued until it is free from air bubbles trapped inside. Next, a minute amount (not more than a small crystal) of the substance being tested (placed on watch glass) is transferred to the borax bead by lightly touching the substance with the hot borax bead. If a large quantity of the substance is transferred on to the borax bead, it will be difficult to make subsequent observations. The bead containing the substance is now heated both in oxidizing and reducing flames. The colour of the bead in both types of flames when the bead is hot and cold (i.e., when taken out of the flame) is used for identifying certain metals, specially transition metals.

After the test is complete, you may heat the bead and gently tap the glass rod so that the bead falls off. The Pt wire should be cleaned as before prior to storing or a subsequent test.

Oxidizing flame	Reducing flame	Metal
Green when hot, blue when cold	Colourless when hot, opaque-red when cold	Cu
Dark yellow when hot, green when cold	Green, hot and cold	Cr
Blue, hot and cold	Blue, hot and cold	Co
Reddish-brown when cold	Grey or black and opaque when cold	Ni
Yellow brown or red when hot, yellow when cold	Green, hot and cold	Ir
Violet, hot and cold	Colourless, hot and cold	Mn

Test for ammonium radical

An easy way to find out the presence of ammonium ions in a mixture is by a simple but effective test: a few drops of sodium hydroxide solution are added to the substance and warmed. Ammonia gas is produced which can be detected by its characteristic smell or by the formation of white fumes of NH_4Cl when the cap of a conc. HCl bottle is held to the path of the vapour. Ammonia can also be detected by litmus paper (red to blue) and by Nessler's reagent or paper (pale yellow to brown)

5. Reactions with mineral acids

With dill. H_2SO_4 and conc. H_2SO_4

A small amount of the substance is taken into a test tube and a few drops of dill. H_2SO_4 are added. Depending on the reaction taking place and gases evolved, certain interferences may be drawn.

Procedure: Treat 10-20 mg of the mixture in a small test tube with about 0.3 -0.5 ml of 1M H_2SO_4 and note whether any reaction takes place in the cold. Heat the mixture on a water bath and test for any gas evolved.

Observations	Inference
Colourless gas is evolved. Gas is odorless and produces turbidity when passed into lime water. With prolonged passage of CO ₂ the turbidity disappears.	CO ₂ from CO ₃ ²⁻ / HCO ₃ ⁻
Pungent reddish – brown fumes / gas	NO ₂
Yellowish –green gas, bleaches litmus paper	Cl ₂ from OCl ⁻
Colourless gas with suffocating odor. Turns filter paper moistened with acidified K ₂ Cr ₂ O ₇ green.	SO ₂ from SO ₃ ⁻
Colourless, odor of foul/rotten eggs gas, which turns filter paper moistened with lead acetate black: Cadmium acetate paper turns yellow.	H ₂ S from S ²⁻

Treat 5-10 mg of the sample in the test tube with 0.3 – 0.5 ml Conc. H₂SO₄. If no reaction occurs in the cold, place the tube in hot water bath.

Observation	Inference
Colourless gas evolved with pungent odor and which fumes in the air: white fumes of NH_4Cl in contact with glass rod moistened with conc. NH_3	HCl from Cl^- $\text{Cl}^- \longrightarrow \text{HCl} + \text{HSO}_4^-$
Gas evolved with pungent odor: reddish colour and fumes in moist air. On adding of MnO_2 increased amount of red fumes with odor of Br_2	HBr and Br_2 from Br^- $\text{Br}^- + \text{H}_2\text{SO}_4 \longrightarrow \text{HBr} + \text{HSO}_4^-$ $\text{H}_2\text{S} + 2\text{HBr} \longrightarrow 2\text{H}_2\text{O} + \text{SO}_2 + \text{Br}_2$
Violet vapors evolved, accompanied by pungent odor.	HI and I_2 from I^- $2\text{I}^- + 2\text{H}_2\text{SO}_4 \longrightarrow \text{I}_2 + 2\text{SO}_4^{2-} + 2\text{H}_2\text{O}$ $\text{I}^- + \text{H}_2\text{SO}_4 \longrightarrow \text{HI} + \text{HSO}_4^-$ $6\text{I}^- + 4\text{H}_2\text{SO}_4 \longrightarrow 3\text{I}_2 + \text{S} + 3\text{SO}_4^{2-} + 4\text{H}_2\text{O}$ $8\text{I}^- + 5\text{H}_2\text{SO}_4 \longrightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{SO}_4^{2-} + 4\text{H}_2\text{O}$
Pungent, brown colored gas evolved. Colour depends upon addition of Cu Turnings (if nitrates absent)	HNO_2 and NO_2 from NO_3^- $4\text{NO}_3^- + 2\text{H}_2\text{SO}_4 \longrightarrow 4\text{NO}_2 + \text{O}_2 + 2\text{SO}_4^{2-} + 2\text{H}_2\text{O}$

References

Vogel's Qualitative Inorganic Analysis 15-20, 37-46, 262-267